

BANA 4373: Advanced Business Analytics for Economics and Business

Lecture 4: Data Preparation II — Merging and Reshaping Data

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Agenda (and where we are in the weekly schedule)

This lecture corresponds to **Week 4: Data Preparation II — Merging and Reshaping**.

Today we will:

- Connect data cleaning (Lecture 3) to combining multiple datasets.
- Learn how economists use **keys** to merge data correctly.
- Understand different types of joins and when to use them.
- Diagnose common merge problems before they affect results.
- Introduce **wide vs long** data structures.

Big picture: Real analysis almost always requires combining data from multiple sources.

How Lecture 4 builds on Lecture 3

Last time, we focused on:

- Cleaning a single dataset
- Fixing types and missing values
- Creating validity flags
- Making work reproducible

Today's shift:

- We move from **one dataset** to **multiple datasets**.
- Errors now come not just from dirty data, but from **incorrect merges**.

Key idea: A perfectly clean dataset can still produce wrong answers if it is merged incorrectly.

Data structures in applied economics

In practice, economists work with three main data structures:

- **Cross-sectional data:**

- Many units, one point in time
- Example: counties in 2023

- **Time series data:**

- One unit, many time periods
- Example: U.S. unemployment over time

- **Panel data:**

- Many units, many time periods
- Example: counties by year

Most applied work aims to construct **panel data**. Merging is how we get there.

Keys: the foundation of merging

A **key** is a variable (or set of variables) that uniquely identifies an observation.

Examples:

- State-level data: `state_fips`
- County-level data: `state_fips + county_fips`
- County-by-year data: `state + county + year`

Why keys matter:

- Merges depend entirely on keys
- If keys are wrong, results are wrong — often silently

Rule: Never merge data until you can clearly state what uniquely identifies a row.

Why names are dangerous keys

Using names (e.g., county or city names) as keys is risky.

Problems:

- Names repeat (e.g., “Washington County”)
- Spelling and formatting differ
- Capitalization and accents vary

Best practice:

- Use numeric or coded identifiers (FIPS)
- Store them as **strings** to preserve leading zeros

This is why Lecture 3 emphasized types and identifiers.

Types of joins (conceptual overview)

When merging two datasets, you must choose how observations are matched.

- **Left join:**
 - Keep all rows from the left dataset
 - Add matching information from the right
- **Inner join:**
 - Keep only rows that appear in both datasets
- **Right join:**
 - Keep all rows from the right dataset
- **Outer join:**
 - Keep all rows from both datasets

The choice of join reflects an **analytical decision**.

Choosing the right join in economics

Different research goals imply different joins.

- **Left join:**

- Most common in applied work
- “This is my main sample; add extra variables if available”

- **Inner join:**

- Useful for balanced samples
- Can silently drop observations

- **Outer join:**

- Useful for diagnostics
- Rarely used for final analysis

Rule: Always check how many rows you gain or lose after a merge.

Common merge failures (and why they matter)

Most merge problems fall into a few categories:

- **Duplicate keys:** creates many-to-many merges
- **Mismatched types:** string vs numeric IDs
- **Wrong join direction:** dropping observations unintentionally
- **Silent row loss:** inner joins remove data without warning

Professional habit:

- Check row counts before and after
- Check missingness after merges
- Use merge diagnostics

Merge diagnostics you should always run

After every merge, ask:

- Did the number of rows change?
- Did new missing values appear?
- Are keys still unique?

In practice, this means:

- `.shape` before and after
- `isna().sum()`
- `duplicated().sum()`

Rule: Never trust a merge you haven't checked.

Wide vs long data

The same data can be stored in different shapes.

Wide format:

- One row per unit
- Many columns (e.g., income_2020, income_2021)

Long format:

- One row per unit–time combination
- Variables like year and income

Most statistical models and plots expect **long data**.

Why reshaping matters

Reshaping is not cosmetic — it enables analysis.

- Panel regressions require long data
- Time trends are easier to plot in long format
- Group operations depend on clear unit–time structure

You will often:

- Merge in wide format
- Reshape to long format for analysis

Mini case (in-class)

We will merge county-level datasets and check:

- Are keys unique before merging?
- How many rows change after each join?
- Where do missing values appear?

Then we will:

- Create a simple county–year structure
- Reshape data from wide to long

Goal: Learn to catch merge problems early — before they affect results.

Key takeaways

- Merging is an analytical decision, not a mechanical step.
- Keys determine whether merges succeed or fail.
- Always diagnose merges before trusting results.
- Long data structures enable most economic analysis.
- Structure first, model second.

End

Questions?